

# ETC5512: Introduction to Data Collection

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## Learning Objectives

- Identify whether data are experimental or observational

- Delineate the data collection methods
- Logically suggest the population that a sample represents

 **Warning**

This tutorial is on the long side. Don't expect to finish all the exercises in your tutorial. You will likely need to follow up in your own time.

## Before your tutorial

**Work** through the following [startR modules](#) on Tidy Data Analysis:

- [Reading data from files](#)
- [Tidy data and restructuring](#)
- [Data manipulation](#)

These should take you ~ 1 hour and 30 minutes.

## Package Installation

Ensure you have the packages installed from Week 1's tutorial.

Also install

```
install.packages("tibble")
install.packages("maps")
install.packages("ggthemes")
```

## Exercise 1

This question relates to the [Tidy Tuesday Data on locations of alternative fuel recharging stations](#). Have a read through this site, and also visit the link to the data providers, DOT.

### a. About the data

Read the details about the data at [DOT](#). How is this data collected, do you think?

## b. Data type

What type of data is this? (observational, experimental, survey, census)

## c. Population vs sample

Describe the population, and what is the sample.

## d. Download and plot the data

Download the data and plot the fueling locations on a map, coloured by fuel type.

```
library(tidyverse)
library(ggthemes)
library(maps)

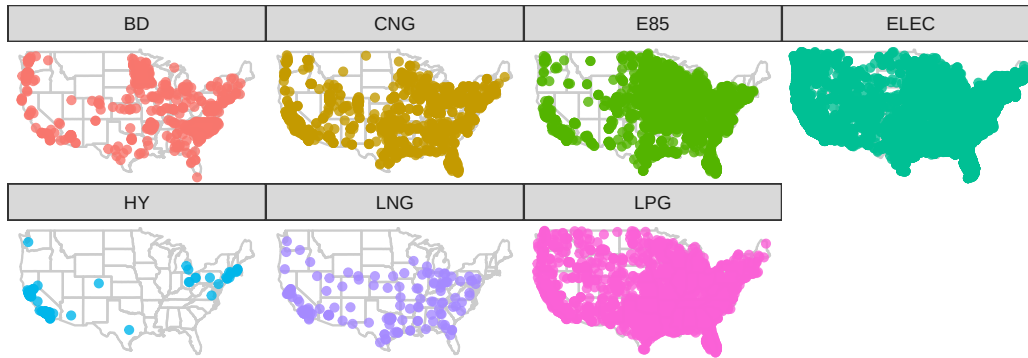
# Note you can read data directly from a website
stations <- read_csv('https://raw.githubusercontent.com/rfordatascience/tidytuesday/master/data/2020/2020-01-13/stations.csv')

# Get the map data for the USA
usa <- map_data("state")

# Filter to continental USA using map boundary
stations <- stations |>
  filter(between(LONGITUDE, min(usa$long)-1, max(usa$long)+1),
         between(LATITUDE, min(usa$lat)-1, max(usa$lat)+1))

# Plot the sites on a map
# Create a different map for each fuel type

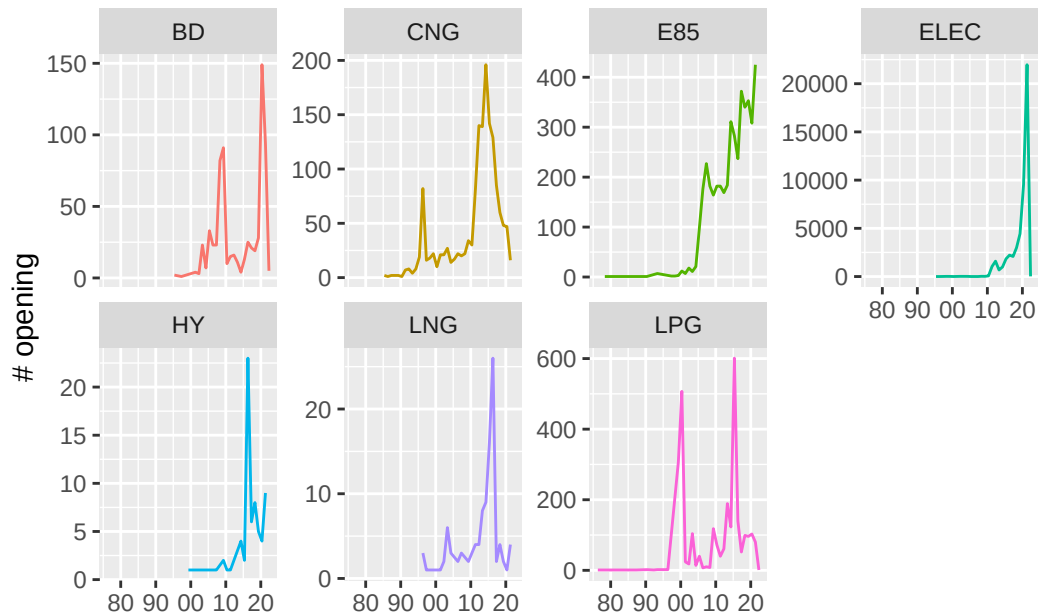
ggplot() +
  geom_path(data=usa, aes(x=long, y=lat, group=group), colour="grey80") +
  geom_point(data=stations, aes(x=LONGITUDE,
                                y=LATITUDE,
                                colour=FUEL_TYPE_CODE),
            alpha=0.8) +
  facet_wrap(~FUEL_TYPE_CODE, ncol=4) +
  coord_map() +
  theme_map() +
  theme(legend.position = "none")
```



### e. New Fuel Stations

Count the number of new stations by month, and make a time series plot by fuel type.

```
# Time line of opening
stations |>
  mutate(OPEN_DATE = as.Date(OPEN_DATE)) |>
  filter(!is.na(OPEN_DATE)) |>
  mutate(m = month(OPEN_DATE),
         yr = year(OPEN_DATE)) |>
  mutate(open_yrmth = as.Date(paste(yr, m, "01", sep="-"), "%Y-%M-%d")) |>
  group_by(open_yrmth, FUEL_TYPE_CODE) |>
  summarise(nopen = n(), .groups = "drop") |>
ggplot(aes(x=open_yrmth,
           y=nopen,
           colour=FUEL_TYPE_CODE)) +
  geom_line() +
  facet_wrap(~FUEL_TYPE_CODE, ncol=4, scales="free_y") +
  ylab("# opening") +
  scale_x_date("", date_labels="%y") +
  theme(legend.position = "none")
```



### g. Fuel growth

If the question to answer is “which alternative fuel vehicle is the fastest growing?” what is the explanatory (independent, predictor) variable and what is the response variable?

## Exercise 2

Here we will look at the [Chocolate bar ratings](#). Details (brief) of how the data was collected are provided [here](#) and more about the data itself is [here](#).

```
chocolate <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tidytuesday/master/data/2020/07/chocolate.csv')
```

### a. Type of data

What type of data is this? (observational, experimental, survey, census)

### b. Data collection

How is the data collected?

### c. Population vs sample

Describe the population.

### d. Response and predictor variables

For the question “Which country of origin of the bean obtains the best rating?” state the response and predictor variables.

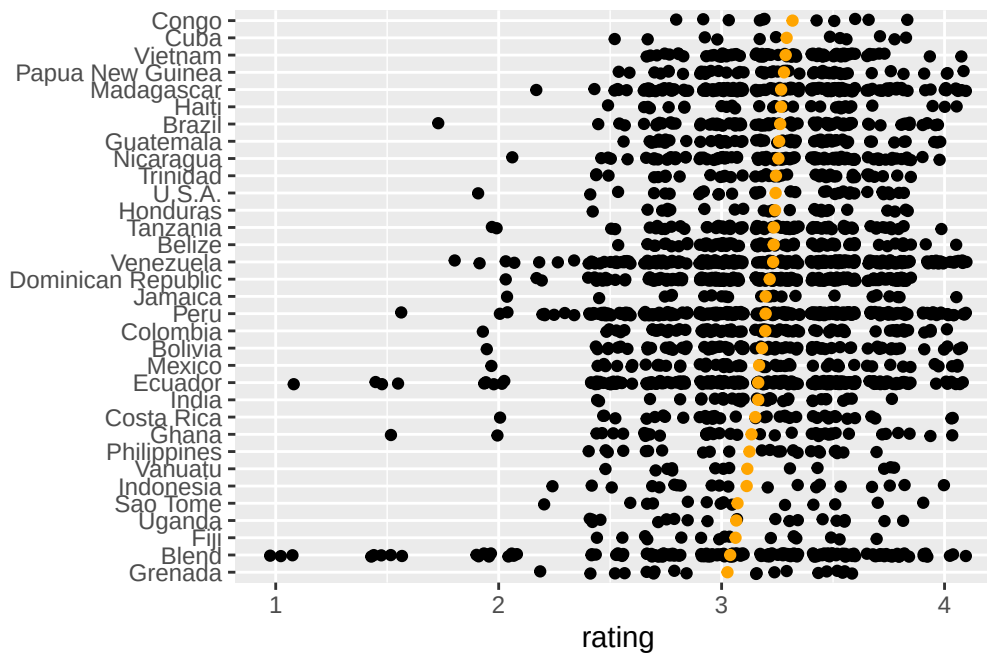
### e. Visualise your data

Make a plot to answer the previous question. (Only use countries with more than 10 records.)

```
# Which countries have more than 10
keep <- chocolate |>
  count(country_of_bean_origin, sort = TRUE) |>
  filter(n>10) |>
  pull(country_of_bean_origin)

# Filter to those countries
chocolate_countries = chocolate |>
  filter(country_of_bean_origin %in% keep)

# Plot the results
ggplot(data = chocolate_countries, aes(x=fct_reorder(country_of_bean_origin, rating, mean)
  y=rating)) +
  geom_jitter(width=0.1) +
  stat_summary(fun = mean, fun.min = median, fun.max = median,
    geom = "point", colour = "orange") +
  xlab("") +
  coord_flip()
```



## In your own time

Read the description of the study titled [“Clearing the Fog: Is Hydroxychloroquine Effective in Reducing COVID-19 Progression \(COVID-19\)”](#).

### a. Data type

What type of data is this? (observational, experimental, survey, census)

### b. Study Participants

How many subjects participated in the study at the start, and to completion?

### c. Study details

What are the:

- experimental units?
- factor?
- blocking factors?
- response variable (outcome measures)?

#### d. Study desing

How are subjects assigned to treatment groups?

#### e. Study results

What were the results of the study?

#### f. Analyse results

Construct the data from the results reported. Compute the proportion of subjects with progression of COVID after 5 days, for the two treatments. Include the standard error of the estimate.

```
hcq <- tibble(trt = c("standard", "standard", "hcq", "hcq"),
              progression = c("all", "yes", "all", "yes"),
              count = c(151, 5, 349, 11))
hcq |>
  pivot_wider(names_from = "progression", values_from = "count") |>
  mutate(p = yes/all) |>
  mutate(se = sqrt(p*(1-p)/all))
```

```
# A tibble: 2 x 5
  trt      all  yes    p    se
<chr> <dbl> <dbl> <dbl> <dbl>
1 standard  151     5 0.0331 0.0146
2 hcq      349    11 0.0315 0.00935
```

#### g. Study conculstions

Based on the proportions and their standard errors, why would the result of the study be that HCQ does NOT improve the outcomes of COVID patients?

#### h. Population vs sample

What is the population for this experiment?

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